

Angry Birds – 2 Player

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1 Introduction

This project implements a two-player turn-based Angry Birds-style game in Pygame. Each player takes turns launching birds from a catapult to destroy the opponent's fortress. Features include six themes (Default, Space, Lava, Desert, Castle, Farm), dynamic wind, theme-based gravity, and destructible blocks (wood, ice, stone) alongside indestructible iron blocks.

2 Modules Used

- **pygame-ce**: For game window management, rendering, sound, and event handling, graphics, sound, and collision detection.
- **numpy**: Assists in mathematical operations like the manipulation of arrays.
- **random**: Simulates randomness in wind speed, direction, and fortress generation.
- **math**: Used for physics calculations such as wind, angle, velocity, and distance.

3 Directory Structure

- **main.py** – entry point, game loop, state management
- **modules/**
 - **bird.py** – Bird class, movement, power-ups
 - **block.py** – Block class, health, collision logic
 - **fortress.py** – procedural fortress generation, block-fall logic
 - **input.py** – name entry fields
 - **text.py** – text rendering
 - **helper.py** – wind noise, rounded-image drawing
 - **variables.py** – global constants (sizes, speeds, health, themes)
- **media/**
 - **images/** – backgrounds, birds, blocks, buttons, icons
 - **audio/** – music, sound effects
 - **fonts/** – Angry Birds font

4 Running Instructions

1. Install dependencies:
`pip install pygame-ce numpy`
2. Launch the game:

```
python main.py
```

3. Follow on-screen prompts:

- Enter player names
- Select theme
- Select difficulty

4. In-game controls:

- Drag & release mouse: launch bird
- **P**: pause/resume
- **R**: restart
- **Q**: quit
- **Space/click**: activate bird power
- Click speaker icon: mute/unmute

5 User Interface Screenshots

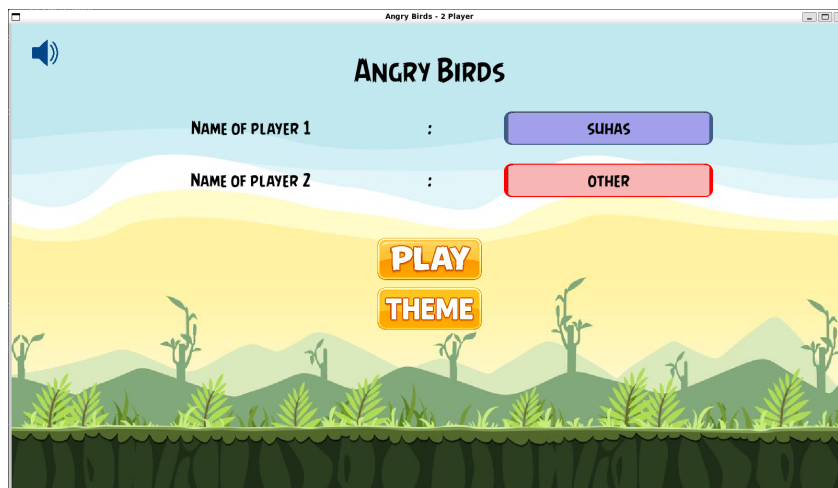


Figure 1: Main menu: enter player names and choose Play or Theme.

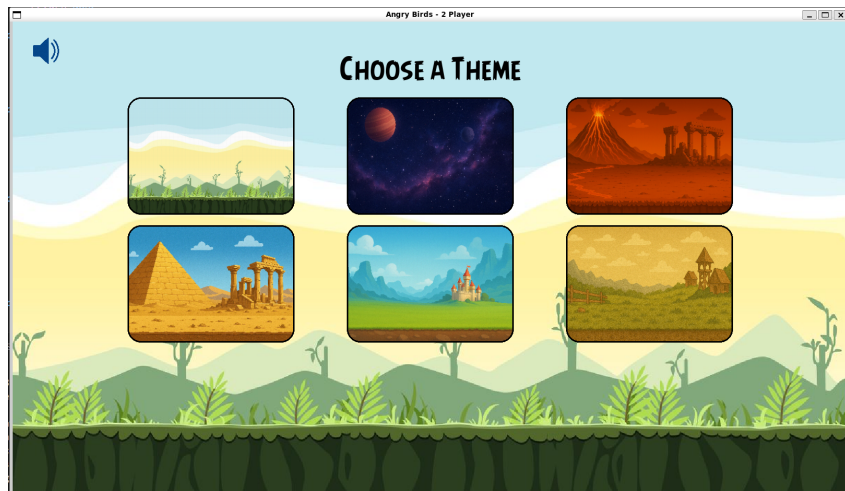


Figure 2: Theme selection screen with six thumbnails.

6 Gameplay Examples

6.1 Space Theme (Zero Gravity)

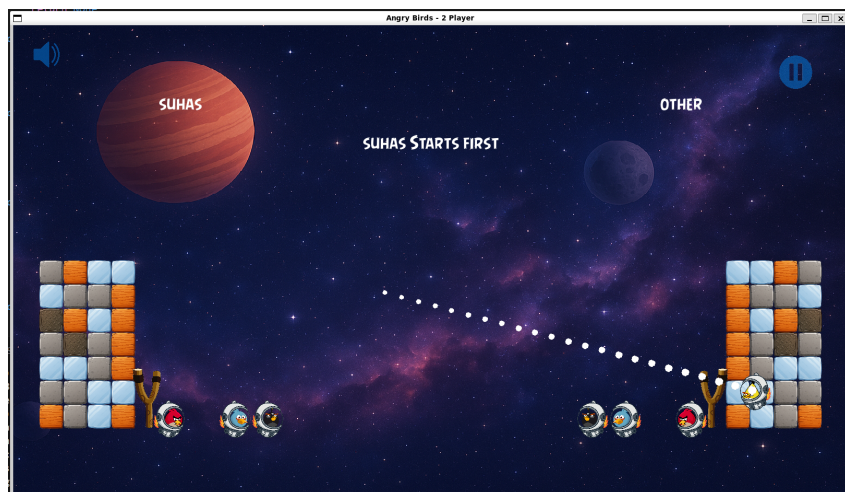


Figure 3: Space theme: gravity disabled, birds fly straight; blocks never fall.

6.2 Default Theme (Normal Gravity)

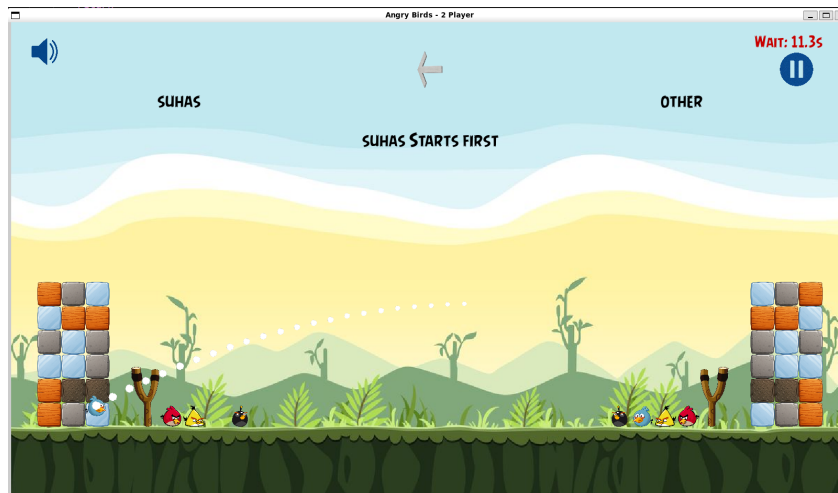


Figure 4: Default theme: gravity applies; wind arrow and countdown timer visible. Wind strength increases with difficulty.

7 Pause and End Screens

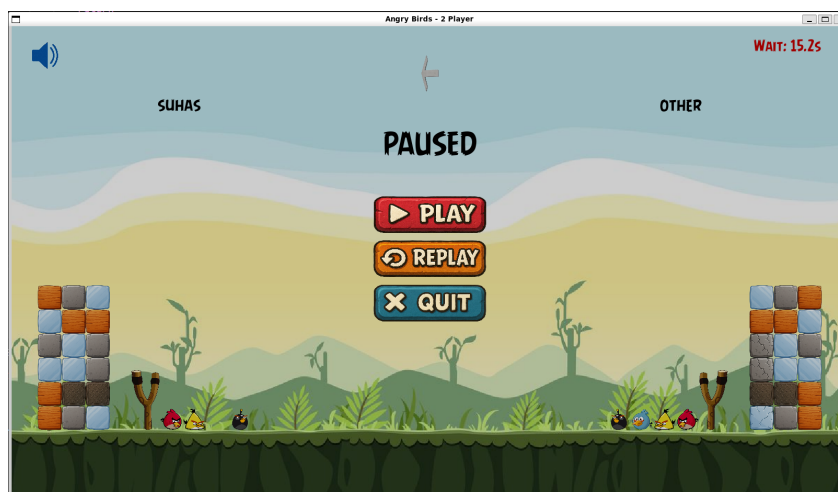


Figure 5: Pause menu: Resume, Replay, Quit.

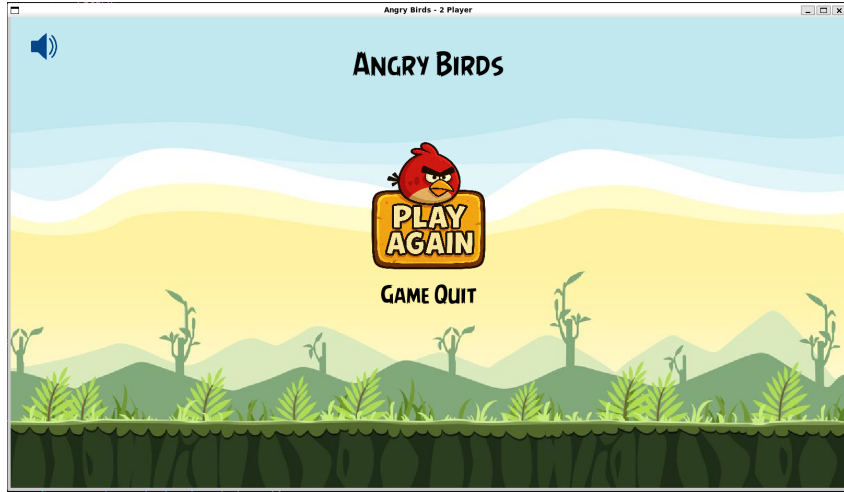


Figure 6: End screen: shows winner or "Game Quit" and Play Again Button.

8 Game Mechanics

8.1 Physics Simulation

Projectile motion is updated for each frame by

$$v'_x = v_x + w(t)^2 \Delta t, \quad v'_y = v_y + g \Delta t,$$

and the new position is

$$x' = x + v_x \Delta t, \quad y' = y + v_y \Delta t,$$

where g is gravity (zero in the Space theme) and $w(t)$ is the noise-based wind term. Collisions invert velocity with a restitution coefficient of 0.3.

8.2 Projectile Generation

Initially each player receives the same four bird types. After those are used, players may choose three additional birds by pressing:

- R : Red bird (Increases in size and hence damage),
- C : Chuck (speed boost),
- B : Blues (splits into three),
- M : Bomb (explodes).

8.3 Wind

Initially, players choose difficulty and theme. After those are selected, if the mode of choice has wind, then the wind is represented with an arrow at the top and a countdown timer (shown next to the wind arrow) limits how long you can wait before launching; if it hits zero, wind becomes constant, and you can no longer wait for wind to become favorable.

9 Basic Features

- Two-player turn-based mechanics
- Four bird types with unique powers
- Destructible blocks: wood, ice, stone
- Symmetric fortress generation for fairness

10 Advanced Features

- Customisations
 - Six themes; space disables gravity entirely.
 - Iron block which stays in place and doesn't get destroyed (in all other themes).
 - Iron blocks are excluded in the Space theme, since zero gravity yields perfectly straight trajectories and blocks behind indestructible iron blocks would be impossible to hit.
- Dynamic wind with maximum strength scaling by difficulty.
- Difficulty settings determine fortress size & wind max.
- Trajectory prediction arc before launch.
- Path of the projectile as bird moves.
- Pause screen and option to restart or quit.
- Projectile effects - Powering up of birds on click.
- Mute/Unmute audio via icon.
- Blocks fall down with gravity.

11 Project Journey

11.1 Key Learnings

- Implementing physics from scratch without external engines.
- Procedural content generation (fortress layout, wind noise).
- Managing multiple game states (menu, theme, difficulty, game, pause, end).
- Modular code organization in Python.
- Balancing features with Time Management.

11.2 Challenges & Solutions

11.2.1 Modularization

As my first large-scale pygame project, separating game logic (`main.py`, `fortress.py`, etc.) from UI (`button.py`, `input.py`) improved maintainability.

11.2.2 Theme Switching

Early theme changes caused inconsistent asset swaps. Introduced new variables to reload backgrounds, fonts, and positions per theme.

11.2.3 Time Constraints

Deadlines limited implementation of extra audio lines and advanced effects. Prioritized core mechanics first to deliver a basic featured product and then extended features.

11.2.4 Lazy asset loading

My first idea was to load images every time any Bird or Block object was created, this caused latency issues when I had to split blue bird into three parts mid air. I solved the problem by loading all the images exactly once before the main loop starts.

12 Game Ideas and References

- Added new ideas to original angry birds: zero-gravity space theme, customizable difficulty, manual bird selection, multiple themes, etc.
- References:
 - Rovio’s angry birds 2 original game
 - Space theme inspired from my interest in astronomy since childhood

13 Bibliography

- Pygame CE Official Documentation: <https://pygame-ce.readthedocs.io/>
- NumPy Reference: <https://numpy.org/doc/>
- Fractal noise implementation in `helper.py` from online sources like reddit and stack overflow
- GeeksforGeeks Pygame Tutorials: <https://www.geeksforgeeks.org/pygame-tutorial/>
- Real Python Pygame Primer: <https://realpython.com/pygame-a-primer/>
- Images: Took images from Angry birds wiki, Shutter-stock, and generated images using ai.
- Angry Birds 2 Trailer <https://www.youtube.com/watch?v=M0niPfYZaaI>
- Wikimedia Commons (public domain game assets): <https://commons.wikimedia.org/>
- Fractal Noise – Wikipedia: https://en.wikipedia.org/wiki/Fractal_noise
- Example projects to get started :<https://github.com/marblexu/PythonAngryBirds> and <https://github.com/marblexu/PythonAngryBirds>